

Module-4 (9 Hours)

Data Collection: Meaning, types, **Data collection methods:** Observations, survey and interview techniques, **Questionnaire design:** Meaning, process of designing questionnaire. Qualitative Techniques of data collection Secondary data Sources: advantages and disadvantages.

Measurement and Scaling Techniques: Basic measurement scales-Nominal scale, Ordinal scale, Interval scale, Ratio scale. Attitude measurement scale - Likert Scale, Semantic Differential Scale, Thurston scale, **Multi-Dimensional Scaling:** Non comparative scaling techniques.

Data Collection:

Meaning

Data is a collection of facts, figures, objects, symbols, and events gathered from different sources. **Organizations collect data with various data collection methods to make better decisions.** Without data, it would be difficult for organizations to make appropriate decisions, so data is collected from different audiences at various points in time.

For instance, an organization must collect data on product demand, customer preferences, and competitors before launching a new product. If data is not collected beforehand, the organization's newly launched product may fail for many reasons, such as less demand and inability to meet customer needs.

Although data is a valuable asset for every organization, it does not serve any purpose until analyzed or processed to get the desired results.

What are Data Collection Methods?

Data collection methods are techniques and procedures used to gather information for research purposes. These methods can range from simple self-reported surveys to more complex experiments and can involve either quantitative or qualitative approaches to data gathering.

Some common data collection methods include surveys, interviews, observations, focus groups, experiments, and secondary data analysis. The data collected through these methods can then be analyzed and used to support or refute research hypotheses and draw conclusions about the study's subject matter.

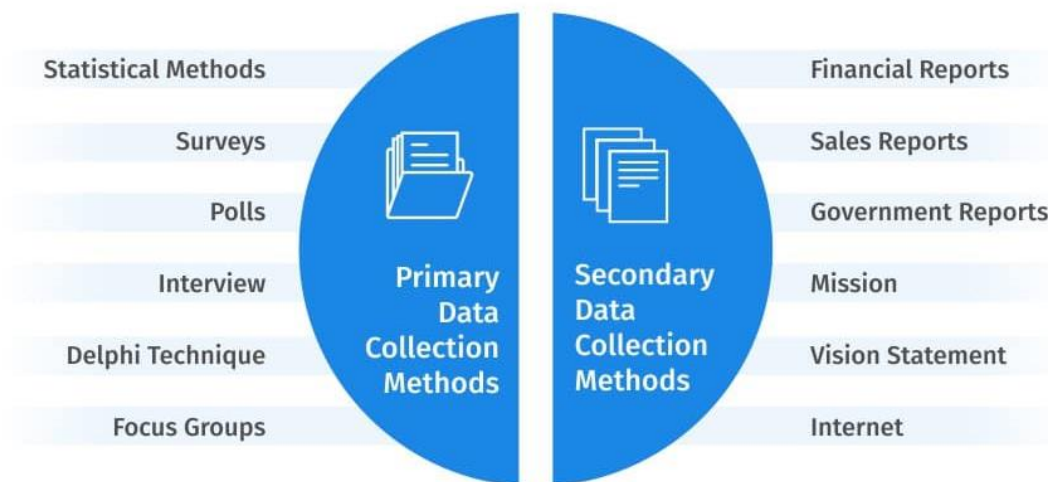
Importance of Data Collection Methods

Data collection methods play a crucial role in the research process as they determine the quality and accuracy of the data collected. Here are some major importance of data collection methods.

- Determines the quality and accuracy of collected data.
- Ensures that the data is relevant, valid, and reliable.
- Helps reduce bias and increase the representativeness of the sample.
- Essential for making informed decisions and accurate conclusions.
- Facilitates achievement of research objectives by providing accurate data.
- Supports the validity and reliability of research findings.

The importance of data collection methods cannot be overstated, as it plays a key role in the overall success and internal validity of the research study.

Types of Data Collection Methods



The choice of data collection method depends on the research question being addressed, the type of data needed, and the resources and time available. You can categorize data collection methods into primary methods of data collection and secondary methods of data collection.

Primary Data Collection Methods

Primary data is collected from first-hand experience and is not used in the past. The data gathered by primary data collection methods are specific to the research's motive and highly accurate.

Primary data collection methods can be divided into two categories: quantitative methods and qualitative methods.

Quantitative Methods:

Quantitative techniques for market research and demand forecasting usually use statistical tools. In these techniques, demand is forecasted based on historical data. These methods of primary data collection are generally used to make long-term forecasts. Statistical methods are highly reliable as subjectivity is minimal in these methods.



Time Series Analysis: The term time series refers to a sequential order of values of a variable, known as a trend, at equal time intervals. Using patterns, an organization can predict the demand for its products and services for the projected time.

Smoothing Techniques: In cases where the time series lacks significant trends, smoothing techniques can be used. They eliminate a random variation from the historical demand. It helps in identifying patterns and demand levels to estimate future demand. The most common methods used in smoothing demand forecasting techniques are the simple moving average method and the weighted moving average method.

Barometric Method: Also known as the leading indicators approach, researchers use this method to speculate future trends based on current developments. When the past events are considered to predict future events, they act as leading indicators.

Qualitative Methods:

Qualitative data collection methods are especially useful in situations when historical data is not available. Or there is no need of numbers or mathematical calculations. Qualitative research is closely associated with words, sounds, feeling, emotions, colors, and other elements that are non-quantifiable. These techniques are based on experience, judgment, intuition, conjecture, emotion, etc.

Quantitative methods do not provide the motive behind participants' responses, often don't reach underrepresented populations, and span long periods to collect the data. Hence, it is best to combine quantitative methods with qualitative methods.

Surveys: Surveys are used to collect data from the target audience and gather insights into their preferences, opinions, choices, and feedback related to their products and services. Most survey software often has a wide range of question types to select.

You can also use a ready-made survey template to save time and effort. Online surveys can be customized as per the business's brand by changing the theme, logo, etc. They can be distributed through several distribution channels such as email, website, offline app, QR code, social media, etc. Depending on the type and source of your audience, you can select the channel.

Once the data is collected, survey software can generate various reports and run analytics algorithms to discover hidden insights. A survey dashboard can give you statistics related to response rate, completion rate, filters based on demographics, export and sharing options, etc. Integrating survey builder with third-party apps can maximize the effort spent on online real-time data collection.

Polls: Polls comprise one single or multiple-choice question. You can go for polls when it is required to have a quick pulse of the audience's sentiments. Because they are short in length, it is easier to get responses from people.

Like surveys, online polls can also be embedded into various platforms. Once the respondents answer the question, they can also be shown how they stand compared to others' responses.

Interviews: In this method, the interviewer asks the respondents face-to-face or by telephone. In face-to-face interviews, the interviewer asks a series of questions to the interviewee in person and notes down responses. If it is not feasible to meet the person, the interviewer can go for a telephone interview. This form of data collection is suitable for only a few respondents. It is too time-consuming and tedious to repeat the same process if there are many participants.

Qualitative Methods



Delphi Technique: In delphi method, market experts are provided with the estimates and assumptions of forecasts made by other experts in the industry. Experts may reconsider and revise their estimates and assumptions based on the information provided by other experts. The consensus of all experts on demand forecasts constitutes the final demand forecast.

Focus Groups: In a focus group, a small group of people, around 8-10 members, discuss the common areas of the research problem. Each individual provides his insights on the issue concerned. A moderator regulates the discussion among the group members. At the end of the discussion, the group reaches a consensus.

Questionnaire: A questionnaire is a printed set of questions, either open-ended or closed-ended. The respondents must answer based on their knowledge and experience with the issue. The questionnaire is a part of the survey, whereas the questionnaire's end goal may or may not be a survey.

Secondary Data Collection Methods

Secondary data is the data that has been used in the past. The researcher can obtain data from the data sources, both internal and external, to the organizational data.

Internal sources of secondary data:

- Organization's health and safety records
- Mission and vision statements
- Financial Statements
- Magazines
- Sales Report
- CRM Software
- Executive summaries

External sources of secondary data:

- Government reports
- Press releases
- Business journals
- Libraries
- Internet

The secondary data collection methods can also involve quantitative and qualitative techniques. Secondary data is easily available and hence, less time-consuming and expensive than primary data. However, with the secondary data collection methods, the authenticity of the data gathered cannot be verified.

The secondary data collection methods can also involve quantitative and qualitative observation techniques. Secondary data is easily available, less time-consuming, and more expensive than primary data. However, with the secondary data collection methods, the authenticity of the data gathered cannot be verified.

Regardless of the data collection method of your choice, there must be direct communication with decision-makers so that they understand and commit to acting according to the results.

For this reason, we must pay special attention to the analysis and presentation of the information obtained. Remember that these data must be useful and functional to us, so the data collection method used has much to do with it.

Advantages of Secondary Research

Secondary research offers a number of advantages to researchers, including efficiency, the ability to build upon existing knowledge, and the ability to conduct research in situations where primary research may not be possible or ethical. By carefully selecting their sources and being thoughtful in their approach, researchers can leverage secondary research to drive impact and advance the field. Some key advantages are the following:

- 1. Most information in this research is readily available.** There are many sources from which relevant data can be collected and used, unlike primary research, where data needs to collect from scratch.
- 2. This is a less expensive and less time-consuming process** as data required is easily available and doesn't cost much if extracted from authentic sources. A minimum expenditure is associated to obtain data.
- 3. The data that is collected through secondary research gives organizations or businesses an idea about the effectiveness of primary research.** Hence, organizations or businesses can form a hypothesis and evaluate cost of conducting primary research.
- 4. Secondary research is quicker to conduct because of the availability of data.** It can be completed within a few weeks depending on the objective of businesses or scale of data needed.

As we can see, this research is the process of analyzing data already collected by someone else, and it can offer a number of benefits to researchers.

Disadvantages of Secondary Research

On the other hand, we have some disadvantages that come with doing secondary research. Some of the most notorious are the following:

1. Although data is readily available, credibility evaluation must be performed to understand the authenticity of the information available.
2. Not all secondary data resources offer the latest reports and statistics. Even when the data is accurate, it may not be updated enough to accommodate recent timelines.

3. Secondary research derives its conclusion from collective primary research data. The success of your research will depend, to a greater extent, on the quality of research already conducted by primary research.

Observation Method

Definition

The observation method is a process that involves human or mechanical observation to observe and describe the behavior of a subject. As the name suggests observational research is a way of collecting relevant information and data by involves observing people's behavior. The observational Research method is also referred to as a participatory study because the researcher has to establish a link with the respondent and for this has to immerse himself in the same setting as theirs. Only then can he use the observational research method to record and take notes.

Meaning and examples

Observational research is used in cases where you want to avoid an error that can be a result of bias during evaluation and interpretation processes. Observational research is one of the many other research methods to obtain objective data by watching a participant and recording it for analysis at a later stage.

A researcher can use observational research, for example in a school, and record the behavior of the children at a young age. Are the children comfortable sharing their tiffin at such an early age will make a good study for the researcher? In this example, the researcher can observe and record the details objectively. The observational research data collection method is associated with observational research methods but with a few ethical issues as it needs the full consent of a research participant.

The observation method in data collection can be

- **Structured observation method** – Structured Observation is a systematic observation method where data is collected as per a pre-defined schedule. The specific variable is used in this observation data collection method.
- **Unstructured observation method** – The unstructured observation method is different from structured observation because it is conducted in a free and open manner without using any pre-determined objectives, schedules, or variables.

Sampling Methods for Observational Data



While conducting observational research, the researcher has a vital role to play in collecting data, as he will have to collect, record and classify the data appropriately. The research methods primary sampling methods are

- **Event sampling** – In the event sampling observation method the researcher decides beforehand what events or behavior he will record and which ones he is going to ignore
- **Time sampling** – In the time sampling observation method, the researcher chooses the time when he will observe. He makes a record of the occurrence only in the specified and pre-determined period
- **Target-time or instantaneous sampling** – In the target-time or instantaneous sampling observation method, the researcher decides beforehand the moments when the observation will happen and will be recorded at that moment. Everything that happens before or after that moment is of no consequence and hence is typically ignored

Types of Observational Research Methods

the different types of observation methods in observational research are as follows-

1) Controlled observations

The controlled observation is carried out in a closed space. It is the researcher who has the authority to decide the place and the time where and when the observation will take place. He also decides who the participants will be and in what circumstances will he use the standardized process.

The participants are chosen for a variable group randomly. The researcher observes and records a detailed and descriptive quantitative data of behavior and divides it into a distinct category. Sometimes the researcher codes the action as per an agreed scale by using a behavior schedule. The coding can include letters or numbers or a range to measure behavior intensity and describe its characteristics.

The collected data from observational research methods is often turned into statistics for laboratory research. In a controlled observation method, the participants are informed by the researcher about the aim of the research. This makes them aware of being observed. The researcher avoids direct contact during the observation method and generally uses a two-way mirror to observe and record details.

2) Naturalistic observations

Social scientists and psychologists generally use the naturalistic observation method in observational research studies. The process of naturalistic observation involves observing and studying the spontaneous behavior of the participants in open or natural surroundings. The role of the researcher is to find and record whatever he can see and observe in natural habitat.

3) Participant observation

The participant observation method is often considered a variant of the naturalistic observation method because it has some similarities with it. The point of difference called disguised naturalistic observation, is that the researcher is not a distant observer anymore because he has joined the participants and become a part of their group. He does this to get a more in-depth and greater insight into their lives.

In participant observation, The researcher interacts with other members of the group freely, participates in their activities, studies their behavior and acquires a different way of life. Participant observation can be overt or covert.

1. **Overt participant observation:** When the researcher asks permission from a group to mingle the observation method is known as overt. He does so by revealing his true purpose and real identity to the group with whom he wants to mingle
2. **Covert participant observation:** When the researcher does not show either his true identity or real meaning to the group he wants to join then the observation is known as covert. He keeps both concealed and takes on a false role and identity to enter and mingle in the group. He generally acts as if he is a genuine member of that group

Advantages of the Observation Method

- Provides direct access to research phenomena
- By observing firsthand, the researcher can collect, check and record accurate data
- Greater flexibility in terms of application
- Generate a permanent record of phenomena and the researcher or others can refer with it later
- The organization method is one of the simplest methods of data collection. It does not require too much technical knowledge
- The observation method is one of the best ways to formulate a hypothesis. The researcher can observe and come to know about the activities, perceptions, likes, and dislikes to form a theory on his subject
- Observational methods are one of the most common methods used in all sciences and is very easy to follow and accept
- In some instances observation is the only available tool to collect essential data and information
- The observation method does not require the willingness of the participant to record. The researcher can observe from a distance and record his findings

Disadvantages of the Observation Method

- Observational methods Face a severe disadvantage because it takes a longer time frame compared to other data collection methods
- There is a chance of higher observer bias in the observation method
- Several personal behaviors are not open for observation and this proves a limitation in case of observation method
- There is a higher chance of the observer influencing the behavior of a sample group elements
- Uncertainties of the event cannot determine the actual time when the event will take place, and this is why every occurrence that is open to observation cannot be observed
- Many of the incidents are abstract like love, affection and the researcher can't gain an exact and correct account of those
- The social phenomena generalization made by observation are not considered reliable as it cannot be used for lab experiments
- In some cases, it is seen that two persons observing the same phenomena come at different results and this can lead to faulty perceptions
- Observation method is considered an expensive affair as it requires hard effort, plenty of time and high cost

Data collection methods: Survey



Survey Data: Definition

Survey data is defined as the resultant data that is collected from a sample of respondents that took a survey. This data is comprehensive information gathered from a target audience about a specific topic to conduct research. There are many methods used for survey data collection and statistical analysis.

Various mediums are used to collect feedback and opinions from the desired sample of individuals. While conducting survey research, researchers prefer multiple sources to gather data such as online surveys, telephonic surveys, face-to-face surveys, etc. The medium of collecting survey data decides the sample of people that are to be reached out to, to reach the requisite number of survey responses.

Factors of collecting survey data such as how the interviewer will contact the respondent (online or offline), how the information is communicated to the respondents etc. decide the effectiveness of gathered data.

Survey Data Collection Methods with Examples

The methods used to collect survey data have evolved with the change in technology. From face-to-face surveys, telephonic surveys to now online and email surveys, the world of survey data collection has changed with time. Each survey data collection method has its pros and cons, and every researcher has a preference for gathering accurate information from the target sample.

The survey response rates for each of these data collection methods will differ as their reach and impact are always different. Different ways are chosen according to specific target population characteristics and the intent to examine human nature under various situations.

Types of survey data based on deployment methods:

There are four main survey data collection methods – Telephonic Surveys, Face-to-face Surveys, and Online Surveys.

- **Online Surveys**

Online surveys are the most cost-effective and can reach the maximum number of people in comparison to the other mediums. The performance of these surveys is much more widespread than the other data collection methods. In situations where there is more than one question to be asked to the target sample, certain researchers prefer conducting online surveys over the traditional face-to-face or telephone surveys.

Online surveys are effective and therefore require computational logic and branching technologies for exponentially more accurate survey data collection vs any other traditional means of surveying. They are straightforward in their implementation and take a minimum time of the respondents. The investment required for survey data collection using online surveys is also negligible in comparison to the other methods. The results are collected in real-time for researchers to analyze and decide corrective measures.

A very good example of an online survey is a hotel chain using an online survey to collect guest satisfaction metrics after a stay or an event at the property.

Online surveys are safe and secure to conduct. As there is no in-person interaction or any direct form of communication, they are quite useful in times of global crisis. For instance, many organizations moved to contactless surveys during the pandemic. It helped them ensure that the employees are not experiencing any COVID-19 symptoms before they come to the office.

- **Face-to-face Surveys**

Gaining information from respondents via face-to-face mediums is much more effective than the other mediums because respondents usually tend to trust the surveyors and provide honest and clear feedback about the subject in-hand.

Researchers can easily identify whether their respondents are uncomfortable with the asked questions and can be extremely productive in case there are sensitive topics involved in the discussion. This online data collection method demands more cost-investment than in comparison to the other methods. According to the geographic segmentation or psychographic segmentation, researchers must be trained to gain accurate information.

For example, a job evaluation survey is conducted in person between an HR or a manager with the employee. This method works best face-to-face as the data collection can collect as accurate information as possible.

- **Telephone Surveys**

Telephone surveys require much lesser investment than face-to-face surveys. Depending on the required reach, telephone surveys cost as much or a little more than online surveys. Contacting respondents via the telephonic medium requires less effort and manpower than the face-to-face survey medium.

If interviewers are located at the same place, they can cross-check their questions to ensure error-free questions are asked to the target audience. The main drawback of conducting telephone surveys is that establishing a friendly equation with the respondent becomes challenging due to the bridge of the medium. Respondents are also highly likely to choose to remain anonymous in their feedback over the phone as the reliability associated with the researcher can be questioned.

For example, if a retail giant would like to understand purchasing decisions, they can conduct a telephonic, motivation, and buying experience survey to collect data about the entire purchasing experience.

- **Paper Surveys**

The other commonly used survey method is paper surveys. These surveys can be used where laptops, computers, and tablets cannot go, and hence they use the age-old method of data collection; pen and paper. This method helps collect survey data in field research and helps strengthen the number of responses collected and the validity of these responses.

A popular example or use case of a paper survey is a fast food restaurant survey where the fast-food chain would like to collect feedback on the dining experience of its patrons.

Types of survey data based on the frequency at which they are administered:

Surveys can be divided into 3 distinctive types on the basis of the frequency of their distribution. They are:

- **Cross-Sectional Surveys**

Cross-sectional surveys are an observational research method that analyzes data of variables collected at one given point of time across a sample population or a pre-defined subset. The survey data from this method helps the researcher understand what the respondent is feeling at a certain point in time. It helps measure opinions in a particular situation.

For example, if the researcher would like to understand movie rental habits, a survey can be conducted across demographics and geographical locations. The cross-sectional survey, for example, can help understand that males between 21-28 rent action movies and females between 35-45 rent romantic comedies. This survey data helps for the basis of a longitudinal study.

- **Longitudinal Surveys**

Longitudinal surveys are those surveys that help researchers to make an observation and collect data over an extended period of time. This survey data can be qualitative or quantitative in nature, and the survey creator does not interfere with the survey respondents.

For example, a longitudinal study can be carried out for years to help understand if mine workers are more prone to lung diseases. This study takes a year and discounts any pre-existing conditions.

- **Retrospective Surveys**

In retrospective surveys, researchers ask respondents to report events from the past. This survey method offers in-depth survey data but doesn't take as long to complete. By deploying this kind of survey, researchers can gather data based on past experiences and beliefs of people.

For example, if hikers are asked about a certain hike – the conditions of the hiking trail, ease of hike, weather conditions, trekking conditions, etc. after they have completed the trek, it is a retrospective study.

Survey Data Analysis

After the survey data has been collected, this data has to be analyzed to ensure it aids towards the end research objective. There are different ways of conducting this research and some steps to follow. They are as below:

Survey Data Analysis: Steps and Tips

There are four main steps of survey data analysis:

1. **Understand the most popular survey research questions:** The survey questions should align with the overall purpose of the survey. That is when the collected data will be effective in helping researchers. For example, if a seminar has been conducted, the researchers will send out a post-seminar feedback survey. The primary goal of this survey will be to understand whether the attendees are interested in attending future seminars. The question will be: “How likely are you to attend future seminars?” – Data collected for this question will decide the likelihood of success of future seminars.
2. **Filter obtained results using the cross-tabulation technique:** Understand the various categories in the target audience and their thoughts using cross-tabulation format. For example, if there are business owners, administrators, students, etc. who attend the seminar, the data about whether they would prefer attending future seminars or not can be represented using cross-tabulation.
3. **Evaluate the derived numbers:** Analyzing the gathered information is critical. How many of the attendees are of the opinion that they will be attending future seminars and how many will not – these facts need to be evaluated according to the results obtained from the sample.
4. **Draw conclusions:** Weave a story with the collected and analyzed data. What was the intention of the survey research, and how does the survey data suffice that objective? – Understand that and develop accurate, conclusive results.

Survey Data Analysis Methods

Conducting a survey without having access to the resultant data and the inability to drawing conclusions from the survey data is pointless. When you conduct a survey, it is imperative to have access to its analytics. It is tough to analyze using traditional survey methods like pen and paper and also requires additional manpower. Survey data analysis becomes much easier when using advanced online data collection methods with an online survey platform such as market research survey software or customer survey software.

Statistical analysis can be conducted on the survey data to make sense of the data that has been collected. There are multiple data analysis methods of quantitative data. Some of the commonly used types are:

- **Cross-tabulation:** Cross-tabulation is the most widely used data analysis methods. It uses a basic tabulation framework to make sense of data. This statistical analysis method helps tabulate data into easily understandable rows and columns, and this helps draw parallels between different research parameters. It contains data that is mutually exclusive or have some connection with each other.
- **Trend analysis:** Trend analysis is a statistical analysis method that provides the ability to look at survey-data over a long period of time. This method helps plot aggregated response data over time allows drawing a trend line of the change, if any, of perceptions over time about a common variable.
- **MaxDiff analysis:** The MaxDiff analysis method is used to gauge what a customer prefers in a product or a service across multiple parameters. For example, a product's feature list, the difference with the competition, ease of use and likert scale, pricing, etc. form the basis for maxdiff analysis. In a simplistic form, this method is also called the "best-worst" method. This method is very similar to conjoint analysis, but it is much easier to implement and can be interchangeably used.
- **Conjoint analysis:** As mentioned above, conjoint analysis is similar to maxdiff analysis, only differing in its complexity and the ability to collect and analyze advance survey data. This method analyzes each parameter behind a person's purchasing behavior. By using conjoint analysis, it is possible to understand what exactly is important to a customer and the aspects that are evaluated before purchase.
- **TURF analysis:** TURF analysis or Total Unduplicated Reach and Frequency analysis, is a statistical research methodology that assesses the total market reach of a product or service or a mix of both. This method is used by organizations to understand the frequency and the avenues at which their messaging reaches customers and prospective customers. This helps them tweak their go-to-market strategies.
- **Gap analysis:** Gap analysis uses a side-by-side matrix question type that helps measure the difference between expected performance and actual performance. This statistical method for survey data helps understand the things that have to be done to move performance from actual to planned performance.
- **SWOT analysis:** SWOT analysis, another widely used statistical method, organizes survey data into data that represents the strength, weaknesses, opportunities, and threats of an organization or product or service that provides a holistic picture of competition. This method helps to create effective business strategies.
- **Text analysis:** Text analysis is an advanced statistical method where intelligent tools make sense of and quantify or fashion qualitative and open-ended data into easily understandable data. This method is used when the survey data is unstructured.

Data Collection Method –

Interview techniques

There are more types of interviews than most people think. An interview is generally a qualitative research technique that involves asking open-ended questions to converse with respondents and collect elicited data about a subject.

The interviewer, in most cases, is the subject matter expert who intends to understand respondent opinions in a well-planned and executed series of star questions and answers.

Interviews are similar to focus groups and surveys for garnering information from the target market but are entirely different in their operation – focus groups are restricted to a small group of 6-10 individuals, whereas surveys are quantitative.

Interviews are conducted with a sample from a population, and the key characteristic they exhibit is their conversational tone.

What is an Interview?

An interview is a way to get information from a person by asking questions and hearing their answers.

An interview is a question-and-answer session where one person asks questions, and the other person answers those questions. It can be a one-on-one, two-way conversation, or there can be more than one interviewer and more than one participant.

The interview is the most important part of the whole selection process. It is used to decide if a person should be interviewed further, hired, or taken out of consideration. It is the main way to learn more about applicants and the basis for judging their job-related knowledge, research skills, and abilities.

Fundamental Types of Interviews in Research

A researcher has to conduct interviews with a group of participants at a juncture in the research where information can only be obtained by meeting and personally connecting with a section of their target audience. Interviews offer the researchers a platform to prompt their participants and obtain inputs in the desired detail. There are three fundamental types of interviews in research:



1. Structured Interviews:

Structured interviews are defined as research tools that could be more flexible in their operations are allows more or no scope of prompting the participants to obtain and analyze results. It is thus also known as a standardized interview and is significantly quantitative in its approach.

Questions in this interview are pre-decided according to the required detail of information. This can be used in a focus group interview and an in-person interview.

These interviews are excessively used in survey research with the intention of maintaining uniformity throughout all the interview sessions.

They can be closed-ended and open-ended – according to the type of target population. Closed-ended questions can be included to understand user preferences from a collection of answer options. In contrast, open-ended ones can be included to gain details about a particular section in the interview.

Example of a structured interview question:

Here's an example of a structured question for a job interview for a customer service job:

- Can you talk about what it was like to work in customer service?

- How do you deal with an angry or upset customer?
- How do you ensure that the information you give customers is correct?
- Tell us about when you went out of your way to help a customer.
- How do you handle a lot of customers or tasks at once?
- Can you talk about how you've used software or tools for customer service?
- How do you set priorities and use your time well while giving good customer service?
- Can you tell us about when you had to get a customer to calm down?
- How do you deal with a customer who wants something that goes against your company's rules?
- Tell me about a time when you had to deal with a hard customer or coworker.

Advantages of structured interviews:

- It focuses on the accuracy of different responses, due to which extremely organized data can be collected. Different respondents have different types of answers to the same structure of questions – answers obtained can be collectively analyzed.
- They can be used to get in touch with a large sample of the target population.
- The interview procedure is made easy due to the standardization offered by it.
- Replication across multiple samples becomes easy due to the same structure of the interview.
- As the scope of detail is already considered while designing the interview questions, better information can be obtained. The researcher can analyze the research problem comprehensively by asking accurate research questions.
- Since the structure of the interview is fixed, it often generates reliable results and is quick to execute.
- The relationship between the researcher and the respondent is not formal, due to which the researcher can clearly understand the margin of error in case the respondent either degree to be a part of the survey or is just not interested in providing the right information.

Disadvantages of structured interviews:

- The limited scope of assessment of obtained results.
- The accuracy of information overpowers the detail of information.
- Respondents are forced to select from the provided answer options.
- The researcher is expected to always adhere to the list of decided questions, irrespective of how interesting the conversation is turning out to be with the participants.
- A significant amount of time is required for a structured interview.

2. Semi-Structured Types of Interviews:

Semi-structured interviews offer a considerable amount of leeway to the researcher to probe the respondents, along with maintaining a basic interview structure. Even if it is a guided conversation between researchers and interviewees – appreciable flexibility is offered to the researchers. A researcher can be assured that multiple interview rounds will not be required in the presence of structure in this type of research interview.

Keeping the structure in mind, the researcher can follow any idea or take creative advantage of the entire interview. Additional respondent probing is always necessary to garner information for a research study. The best application of semi-structured interviews is when the researcher doesn't have time to conduct research and requires detailed information about the topic.

Example of a semi-structured interview question:

Here's an example of a semi-structured marketing job interviews question:

- Can you tell us about the marketing work you've done?
- What do you think are the most important parts of a marketing campaign that works?
- Tell me about a campaign you worked on that you're very proud of.
- How do you do research on the market and look at data to help you make marketing decisions?
- Can you tell us about a time when you had to change your marketing plan because of something that didn't go as planned?
- How do you figure out if a marketing campaign worked?
- Can you talk about how you've used social media to market?
- How do you ensure your marketing message gets through to the people you want to hear it?
- Can you tell us about a time when you had to run a marketing campaign on a small budget?
- How do you keep up with changes and trends in marketing?

Advantages of semi-structured interviews:

- Questions from semi-structured interview questions are prepared before the scheduled interview, giving the researcher time to prepare and analyze the questions.
- It is flexible to an extent while maintaining the research guidelines.
- Unlike a structured interview, researchers can express the interview questions in the preferred format.
- Reliable qualitative data can be collected via these interviews.
- The flexible structure of the interview.

Disadvantages of semi-structured interviews:

- Participants may question the reliability factor of these interviews due to the flexibility offered.
- Comparing two different answers becomes difficult as the guideline for conducting interviews is not entirely followed. No two questions will have the exact same structure, and the result will be an inability to compare and infer results.

3. Unstructured Interviews:

Also called in-depth interviews, unstructured interviews are usually described as conversations held with a purpose in mind – to gather data about the research study. These interviews have the least number of questions as they lean more towards a normal conversation but with an underlying subject.

The main objective of most researchers using unstructured interviews is to build a bond with the respondents, due to which there is a high chance that the respondents will be 100% truthful with their answers. There are no guidelines for the researchers to follow. So they can approach the participants ethically to gain as much information as possible about their research topic.

Since there are no guidelines for these interviews, a researcher is expected to keep their approach in check so that the respondents do not sway away from the main research motive.

For a researcher to obtain the desired outcome, he/she must keep the following factors in mind:

- The intent of the interview.
- The interview should primarily take into consideration the participant's interests and skills.
- All the conversations should be conducted within the permissible limits of research, and the researcher should try and stick by these limits.
- The researcher's skills and knowledge should match the interview's purpose.
- Researchers should understand the dos and don'ts of it.

Example of an unstructured interview question:

Here's an example of a question asked in an unstructured interview:

- Can you tell me about when you had to deal with something hard and how you did it?
- What are some of the things you're most proud of, and what did you learn from them?
- How do you deal with ambiguity or not knowing what to do at work?
- Can you describe how you lead and how you get your team going?

- Tell me about a time when you had to take a chance and how it turned out.
- What do you think are the most important qualities for success in this role?
- How do you deal with setbacks or failures, and what do you learn from them?
- Can you tell me about a time when you had to solve a problem by thinking outside the box?
- What do you think makes you different from the other people who want this job?
- Can you tell me about a time when you had to make a hard choice and how you made that choice?

Advantages of Unstructured Interviews:

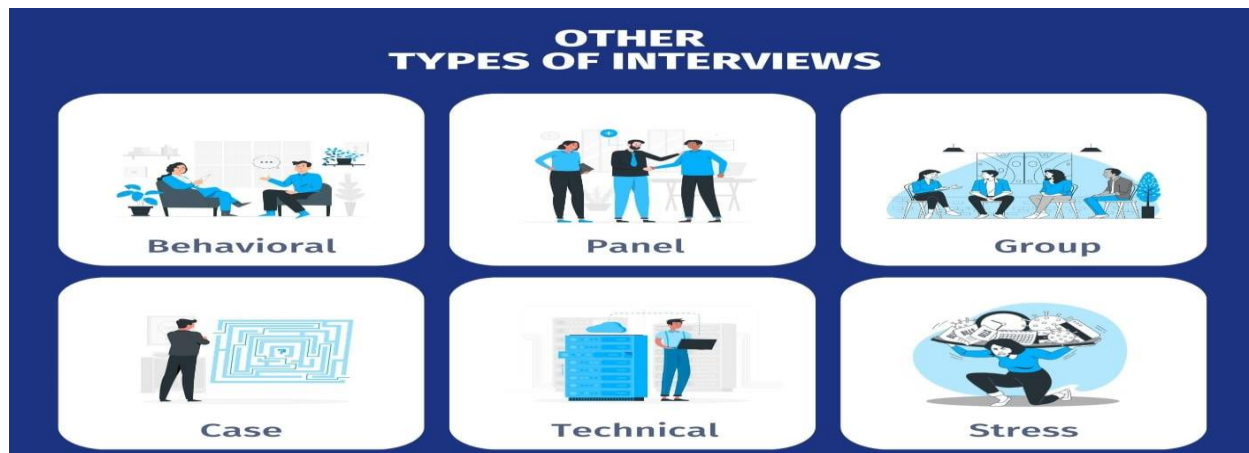
- Due to this type of interview's informal nature, it becomes extremely easy for researchers to try and develop a friendly rapport with the participants. This leads to gaining insights in extreme detail without much conscious effort.
- The participants can clarify all their doubts about the questions, and the researcher can take each opportunity to explain his/her intention for better answers.
- There are no questions that the researcher has to abide by, and this usually increases the flexibility of the entire research process.

Disadvantages of Unstructured Interviews:

- Researchers take time to execute these interviews because there is no structure to the interview process.
- The absence of a standardized set of questions and guidelines indicates that its reliability of it is questionable.
- The ethics involved in these interviews are often considered borderline upsetting.

Other Types of Interviews

Besides the 3 basic interview types, we have already mentioned there are more. Here are some other interview types that are commonly used in a job interview:



Behavioral Interview

During this type of interview, candidates are asked to give specific examples of how they have acted in the past. The idea behind this kind of interview is that what someone did in the past can be a sign of how they will act in the future. And by this interview, the company can also understand the interviewee's behavior through body language.

Panel Interview

During a panel interview, three or more interviewers usually ask questions and evaluate the candidate's answers as a group. This is a good way to get a full picture of a candidate's skills and suitability for the job.

Group Types of Interviews

Multiple people are interviewed at the same time in group interviews. This form of interview often focus groups that are utilized on entry-level positions or employment in customer service to examine how well candidates get along with others and function as a team.

Case Interview

During a case interview, candidates are given a business problem or scenario and asked to think about how to solve it. In the consulting and finance fields, this kind of interview is common.

Technical Interview

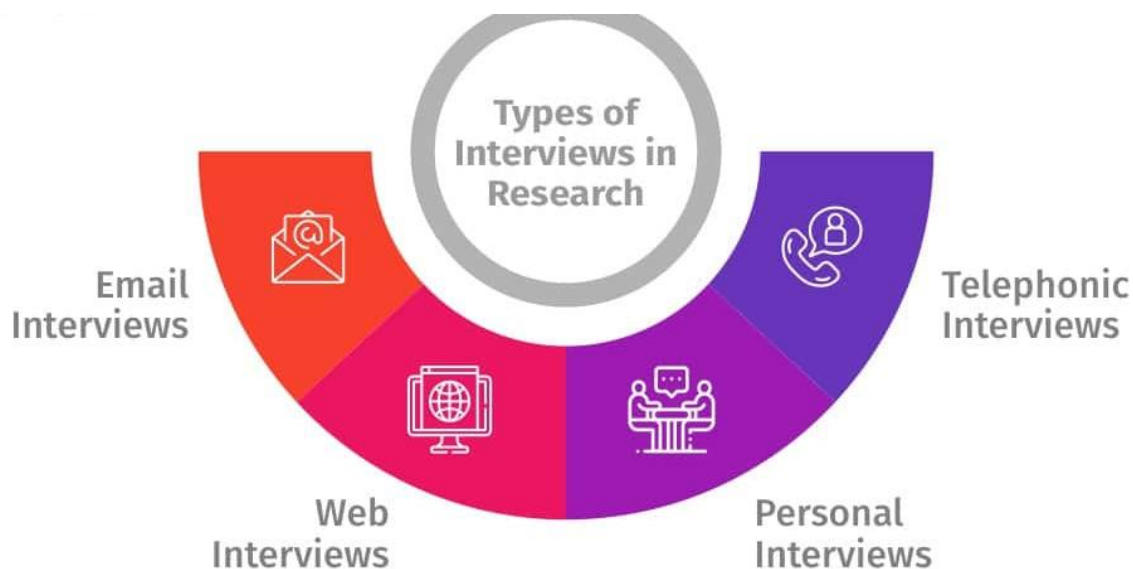
A candidate's technical skills and knowledge are tested during a technical interview, usually in fields like engineering or software development. Most of the time, candidates are asked to solve problems or complete technical tasks.

Stress Interview

During a stress interview, candidates are put under pressure or asked difficult or confrontational questions on purpose to see how they react in stressful situations. This kind of interview is used to see how well a candidate can deal with stress and hard situations.

Methods of Research Interviews:

There are four methods to conduct research interviews, each of which is peculiar in its application and can be used according to the research study requirement.



Personal Interviews:

Personal interviews are one of the most used types of interviews, where the questions are asked personally directly to the respondent as a form of an individual interview. One of the many in-person interviews is a lunch interview, which is frequently better suited for casual inquiries and discussions.

For this, a researcher can have a guide to online surveys to take note of the answers. A researcher can design his/her survey in such a way that they take notes of the comments or points of view that stands out from the interviewee. It can be a one-on-one interview as well.

Advantage:

- Higher response rate.
- When the interviewees and respondents are face-to-face, there is a way to adapt the questions if this is not understood.
- More complete answers can be obtained if there is doubt on both sides or a remarkable piece of information is detected.
- The researcher has an opportunity to detect and analyze the interviewee's body language at the time of asking the questions and taking notes about it.

Disadvantages:

- They are time-consuming and extremely expensive.
- They can generate distrust on the part of the interviewee since they may be self-conscious and not answer truthfully.
- Contacting the interviewees can be a real headache, either scheduling an appointment in workplaces or going from house to house and not finding anyone.
- Therefore, many interviews are conducted in public places like shopping centers or parks. Even consumer studies take advantage of these sites to conduct interviews or surveys and give incentives, gifts, and coupons. In short, There are great opportunities for online research in shopping centers.
- Among the advantages of conducting such types of interviews is that the respondents will have more fresh information if the interview is conducted in the context and with the appropriate stimuli so that researchers can have data from their experience at the scene of the events immediately and first hand. The interviewer can use an online survey through a mobile device that will undoubtedly facilitate the entire process.

Telephonic Type of Interviews:

Phonic interviews are widely used and easily combined with online surveys to conduct research effectively.

Advantages:

- To find the interviewees, it is enough to have their phone numbers on hand.
- They are usually lower cost.
- The information is collected quickly.
- Having a personal contact can also clarify doubts or give more details of the questions.

Disadvantages:

- Many times researchers observe that people do not answer phone calls because it is an unknown number for the respondent or simply already changed their place of residence and they cannot locate it, which causes a bias in the interview.
- Researchers also face that they simply do not want to answer and resort to pretexts such as they are busy to answer, they are sick, they do not have the authority to answer the questions asked, they have no interest in answering, or they are afraid of putting their security at risk.
- One of the aspects that should be taken care of in these types of interviews is the kindness with which the interviewers address the respondents in order to get them to cooperate more easily with their answers. Good communication is vital for the generation of better answers.

Email or Web Page Types of Interviews:

Online research is growing more and more because consumers are migrating to a more virtual world, and it is best for each researcher to adapt to this change.

The increase in people with Internet access has made it popular that interviews via email or web page stand out among the types of interviews most used today. For this nothing better than an online survey.

More and more consumers are turning to online shopping, which is why they are a great niche to be able to carry out an interview that will generate information for the correct decision-making.

Advantages of email surveys:

- Speed in obtaining data
- The respondents respond according to their time, when they want, and where they decide.
- Online surveys can be mixed with other research methods or using some of the previous interview models. They are tools that can perfectly complement and pay for the project.
- A researcher can use a variety of questions and logic to create graphs and reports immediately.

Disadvantages of email survey:

- Low response rates
- Limited access to certain populations
- Potential for spam filters
- Lack of personal touch

Questionnaire design:

What is a Questionnaire?

A questionnaire is a research instrument that consists of a set of questions or other types of prompts that aims to collect information from a respondent. A research questionnaire is typically a mix of close-ended questions and open-ended questions.

Open-ended, long-form questions offer the respondent the ability to elaborate on their thoughts. Research questionnaires were developed in 1838 by the Statistical Society of London.

The data collected from a data collection questionnaire can be both qualitative as well as quantitative in nature. A questionnaire may or may not be delivered in the form of a survey, but a survey always consists of a questionnaire.

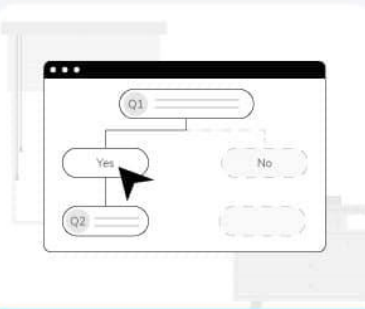
Advantages of a good questionnaire design

- With a survey questionnaire, you can gather a lot of data in less time.
- There is less chance of any bias creeping if you have a standard set of questions to be used for your target audience. You can apply logic to questions based on the respondents' answers, but the questionnaire will remain standard for a group of respondents that fall in the same segment.
- Surveying online survey software is quick and cost-effective. It offers you a rich set of features to design, distribute, and analyze the response data.
- It can be customized to reflect your brand voice. Thus, it can be used to reinforce your brand image.
- The responses can be compared with the historical data and understand the shift in respondents' choices and experiences.
- Respondents can answer the questionnaire without revealing their identity. Also, many survey software complies with significant data security and privacy regulations.

Advantages of a well-designed online questionnaire



Collect a large amount of data in less time.



There is less chance of bias with the help of logic questions.



Online survey software is fast and cost effective to design, distribute and analyze.



It can be customized to reflect your brand voice.



Responses can be compared with historical data.



Respondents can answer the questionnaire without revealing their identity.

Characteristics of a good questionnaire

Your survey design depends on the type of information you need to collect from respondents. Qualitative questionnaires are used when there is a need to collect exploratory information to help prove or disprove a hypothesis. Quantitative questionnaires are used to validate or test a previously generated hypothesis. However, most questionnaires follow some essential characteristics:

- **Uniformity:** Questionnaires are very useful to collect demographic information, personal opinions, facts, or attitudes from respondents. One of the most significant attributes of a research form is uniform design and standardization. Every respondent sees the same questions. This helps in data collection and statistical analysis of this data. For example, the retail store evaluation questionnaire template contains questions for evaluating retail store experiences. Questions relate to purchase value, range of options for product selections, and quality of merchandise. These questions are uniform for all customers.

- **Exploratory:** It should be exploratory to collect qualitative data. There is no restriction on questions that can be in your questionnaire. For example, you use a data collection questionnaire and send it to the female of the household to understand her spending and saving habits relative to the household income. Open-ended questions give you more insight and allow the respondents to explain their practices. A very structured question list could limit the data collection.
- **Question Sequence:** It typically follows a structured flow of questions to increase the number of responses. This sequence of questions is screening questions, warm-up questions, transition questions, skip questions, challenging questions, and classification questions. For example, our motivation and buying experience questionnaire template covers initial demographic questions and then asks for time spent in sections of the store and the rationale behind purchases.

Types & Definitions

As we explored before, questionnaires can be either structured or free-flowing. Let's take a closer look at what that entails for your surveys.

- **Structured Questionnaires:** Structured questionnaires collect quantitative data. The questionnaire is planned and designed to gather precise information. It also initiates a formal inquiry, supplements data, checks previously accumulated data, and helps validate any prior hypothesis.
- **Unstructured Questionnaires:** Unstructured questionnaires collect qualitative data. They use a basic structure and some branching questions but nothing that limits the responses of a respondent. The questions are more open-ended to collect specific data from participants.

Types of questions in a questionnaire

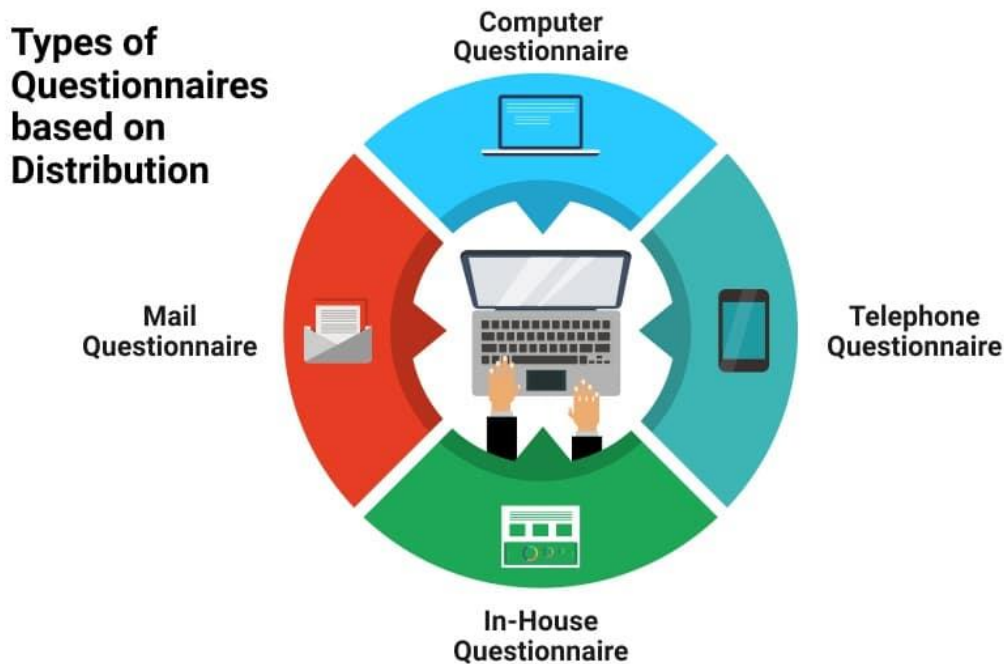
You can use multiple question types in a questionnaire. Using various question types can help increase responses to your research questionnaire as they tend to keep participants more engaged. The best customer satisfaction survey templates are the most commonly used for better insights and decision-making.

Some of the widely used types of questions are:

- **Open-Ended Questions:** Open-ended questions help collect qualitative data in a questionnaire where the respondent can answer in a free form with little to no restrictions.
- **Dichotomous Questions:** The dichotomous question is generally a “yes/no” close-ended question. This question is usually used in case of the need for necessary validation. It is the most natural form of a questionnaire.

- **Multiple-Choice Questions:** Multiple-choice questions are a close-ended question type in which a respondent has to select one (single-select multiple-choice question) or many (multi-select multiple choice question) responses from a given list of options. The multiple-choice question consists of an incomplete stem (question), right answer or answers, incorrect answers, close alternatives, and distractors. Of course, not all multiple-choice questions have all of the answer types. For example, you probably won't have the wrong or right answers if you're looking for customer opinion.
- **Scaling Questions:** These questions are based on the principles of the four measurement scales – nominal, ordinal, interval, and ratio. A few of the question types that utilize these scales' fundamental properties are rank order questions, Likert scale questions, semantic differential scale questions, and Stapel scale questions.
- **Pictorial Questions:** This question type is easy to use and encourages respondents to answer. It works similarly to a multiple-choice question. Respondents are asked a question, and the answer choices are images. This helps respondents choose an answer quickly without overthinking their answers, giving you more accurate data.

Types of Questionnaires



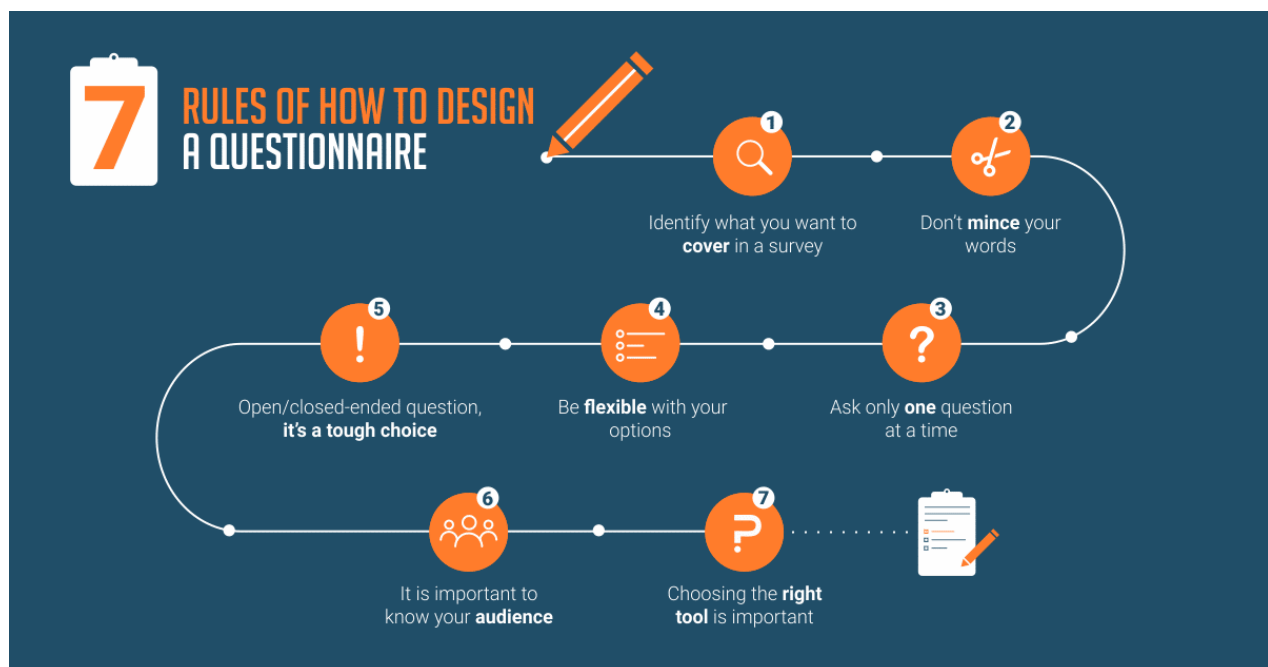
Questionnaires can be administered or distributed in the following forms:

- **Online Questionnaire:** In this type, respondents are sent the questionnaire via email or other online mediums. This method is generally cost-effective and time-efficient. Respondents can

also answer at leisure. Without the pressure to respond immediately, responses may be more accurate. The disadvantage, however, is that respondents can easily ignore these questionnaires. Read more about online surveys.

- **Telephone Questionnaire:** A researcher makes a phone call to a respondent to collect responses directly. Responses are quick once you have a respondent on the phone. However, a lot of times, the respondents hesitate to give out much information over the phone. It is also an expensive way of conducting research. You're usually not able to collect as many responses as other types of questionnaires, so your sample may not represent the broader population.
- **In-House Questionnaire:** This type is used by a researcher who visits the respondent's home or workplace. The advantage of this method is that the respondent is in a comfortable and natural environment, and in-depth data can be collected. The disadvantage, though, is that it is expensive and slow to conduct.
- **Mail Questionnaire:** These are starting to be obsolete but are still being used in some market research studies. This method involves a researcher sending a physical data collection questionnaire request to a respondent that can be filled in and sent back. The advantage of this method is that respondents can complete this on their own time to answer truthfully and entirely. The disadvantage is that this method is expensive and time-consuming. There is also a high risk of not collecting enough responses to make actionable insights from the data.

How to design a Questionnaire



Questionnaire design is a multistep process that requires attention to detail at every step.

Researchers are always hoping that the responses received for a survey questionnaire yield useable data. If the questionnaire is too complicated, there is a fair chance that the respondent might get confused and will drop out or answer inaccurately.

As a survey creator, you may want to pre-test the survey by administering it to a focus group during development. You can try out a few different questionnaire designs to determine which resonates best with your target audience. Pre-testing is a good practice as the survey creator can comprehend the initial stages if there are any changes required in the survey.

Steps Involved in Questionnaire Design

1. Identify the scope of your research:

Think about what your questionnaire is going to include before you start designing the look of it. The clarity of the topic is of utmost importance as this is the primary step in creating the questionnaire. Once you are clear on the purpose of the questionnaire, you can begin the design process.

2. Keep it simple:

The words or phrases you use while writing the questionnaire must be easy to understand. If the questions are unclear, the respondents may simply choose any answer and skew the data you collect.

3. Ask only one question at a time:

At times, a researcher may be tempted to add two similar questions. This might seem like an excellent way to consolidate answers to related issues, but it can confuse your respondents or lead to inaccurate data. If any of your questions contain the word “and,” take another look. This question likely has two parts, which can affect the quality of your data.

4. Be flexible with your options:

While designing, the survey creator needs to be flexible in terms of “option choice” for the respondents. Sometimes the respondents may not necessarily want to choose from the answer options provided by the survey creator. An “other” option often helps keep respondents engaged in the survey.

5. The open-ended or closed-ended question is a tough choice:

The survey creator might end up in a situation where they need to make distinct choices between open or close-ended questions. The question type should be carefully chosen as it defines the tone and importance of asking the question in the first place.

If the questionnaire requires the respondents to elaborate on their thoughts, an open-ended question is the best choice. If the surveyor wants a specific response, then close-ended questions should be their primary choice. The key to asking closed-ended questions is to generate data that is easy to analyze and spot trends.

6. It is essential to know your audience:

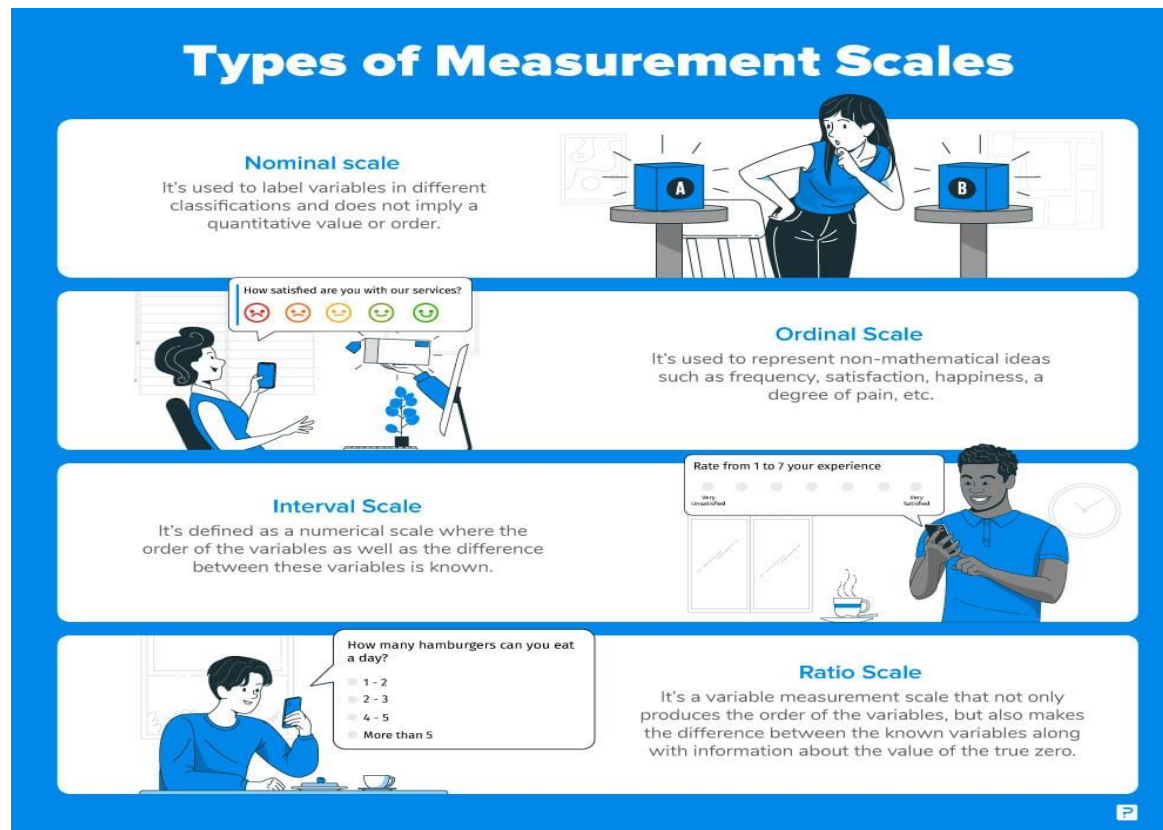
A researcher should know their target audience. For example, if the target audience speaks mostly Spanish, sending the questionnaire in any other language would lower the response rate and accuracy of data. Something that may seem clear to you may be confusing to your respondents. Use simple language and terminology that your respondents will understand, and avoid technical jargon and industry-specific language that might confuse your respondents.

For efficient market research, researchers need a representative sample collected using one of the many sampling techniques, such as a sample questionnaire. It is imperative to plan and define these target respondents based on the demographics required.

7. Choosing the right tool is essential:

Always save personal questions for last. Sensitive questions may cause respondents to drop off before completing. If these questions are at the end, the respondent has had time to become more comfortable with the interview and are more likely to answer personal or demographic questions.

Measurement and Scaling Techniques



Nominal Scale: 1st Level of Measurement

Nominal Scale, also called the categorical variable scale, is defined as a scale that labels variables into distinct classifications and doesn't involve a quantitative value or order. This scale is the simplest of the four variable measurement scales. Calculations done on these variables will be futile as the options have no numerical value.

There are cases where this scale is used for the purpose of classification – the numbers associated with variables of this scale are only tags for categorization or division. Calculations done on these numbers will be futile as they have no quantitative significance.

For a question such as:

Where do you live?

- 1- Suburbs

- 2- City
- 3- Town

Nominal scale is often used in research surveys and questionnaires where only variable labels hold significance.

For instance, a customer survey asking “Which brand of smartphones do you prefer?”

Options : “Apple”- 1 , “Samsung”-2, “OnePlus”-3.

- In this survey question, only the names of the brands are significant for the researcher conducting consumer research or netnography. There is no need for any specific order for these brands. However, while capturing nominal data, researchers conduct analysis based on the associated labels.
- In the above example, when a survey respondent selects Apple as their preferred brand, the data entered and associated will be “1”. This helped in quantifying and answering the final question – How many respondents selected Apple, how many selected Samsung, and how many went for OnePlus – and which one is the highest.
- This is the fundamental of quantitative research, and nominal scale is the most fundamental research scale.

Nominal Scale Data and Analysis

There are two primary ways in which nominal scale data can be collected:

1. By asking an open-ended question, the answers of which can be coded to a respective number of label decided by the researcher.
2. The other alternative to collect nominal data is to include a multiple choice question in which the answers will be labeled.

In both cases, the analysis of gathered data will happen using percentages or mode, i.e., the most common answer received for the question. It is possible for a single question to have more than one mode as it is possible for two common favorites can exist in a target population.

Nominal Scale Examples

- Gender
- Political preferences
- Place of residence

What is your Gender?	What is your Political preference?	Where do you live?
• M- Male • F- Female	• 1- Independent • 2- Democrat • 3- Republican	• 1- Suburbs • 2- City • 3- Town

Ordinal Scale: 2nd Level of Measurement

Ordinal Scale is defined as a variable measurement scale used to simply depict the order of variables and not the difference between each of the variables. These scales are generally used to depict non-mathematical ideas such as frequency, satisfaction, happiness, a degree of pain, etc. It is quite straightforward to remember the implementation of this scale as ‘Ordinal’ sounds similar to ‘Order’, which is exactly the purpose of this scale.

Ordinal Scale maintains descriptive qualities along with an intrinsic order but is void of an origin of scale and thus, the distance between variables can’t be calculated. Descriptive qualities indicate tagging properties similar to the nominal scale, in addition to which, the ordinal scale also has a relative position of variables. Origin of this scale is absent due to which there is no fixed start or “true zero”.

Ordinal Data and Analysis

Ordinal scale data can be presented in tabular or graphical formats for a researcher to conduct a convenient analysis of collected data. Also, methods such as Mann-Whitney U test and Kruskal–Wallis H test can also be used to analyze ordinal data. These methods are generally implemented to compare two or more ordinal groups.

In the Mann-Whitney U test, researchers can conclude which variable of one group is bigger or smaller than another variable of a randomly selected group. While in the Kruskal–Wallis H test, researchers can analyze whether two or more ordinal groups have the same median or not.

Ordinal Scale Examples

Status at workplace, tournament team rankings, order of product quality, and order of agreement or satisfaction are some of the most common examples of the ordinal Scale. These scales are generally used in market research to gather and evaluate relative feedback about product satisfaction, changing perceptions with product upgrades, etc.

For example, a semantic differential scale question such as:

How satisfied are you with our services?

- Very Unsatisfied – 1
- Unsatisfied – 2
- Neutral – 3
- Satisfied – 4
- Very Satisfied – 5

1. Here, the order of variables is of prime importance and so is the labeling. Very unsatisfied will always be worse than unsatisfied and satisfied will be worse than very satisfied.
2. This is where ordinal scale is a step above nominal scale – the order is relevant to the results and so is their naming.
3. Analyzing results based on the order along with the name becomes a convenient process for the researcher.
4. If they intend to obtain more information than what they would collect using a nominal scale, they can use the ordinal scale.

This scale not only assigns values to the variables but also measures the rank or order of the variables, such as:

- Grades
- Satisfaction
- Happiness

How satisfied are you with our services?

- 1- Very Unsatisfied
- 2- Unsatisfied
- 3- Neural
- 4- Satisfied
- 5- Very Satisfied

Interval Scale: 3rd Level of Measurement

Interval Scale is defined as a numerical scale where the order of the variables is known as well as the difference between these variables. Variables that have familiar, constant, and computable differences are classified using the Interval scale. It is easy to remember the primary role of this scale

too, 'Interval' indicates 'distance between two entities', which is what Interval scale helps in achieving.

These scales are effective as they open doors for the statistical analysis of provided data. Mean, median, or mode can be used to calculate the central tendency in this scale. The only drawback of this scale is that there no pre-decided starting point or a true zero value.

Interval scale contains all the properties of the ordinal scale, in addition to which, it offers a calculation of the difference between variables. The main characteristic of this scale is the equidistant difference between objects.

For instance, consider a Celsius/Fahrenheit temperature scale –

- 80 degrees is always higher than 50 degrees and the difference between these two temperatures is the same as the difference between 70 degrees and 40 degrees.
- Also, the value of 0 is arbitrary because negative values of temperature do exist – which makes the Celsius/Fahrenheit temperature scale a classic example of an interval scale.
- Interval scale is often chosen in research cases where the difference between variables is a mandate – which can't be achieved using a nominal or ordinal scale. The Interval scale quantifies the difference between two variables whereas the other two scales are solely capable of associating qualitative observation values with variables.
- The mean and median values in an ordinal scale can be evaluated, unlike the previous two scales.
- In statistics, interval scale is frequently used as a numerical value can not only be assigned to variables but calculation on the basis of those values can also be carried out.

Even if interval scales are amazing, they do not calculate the “true zero” value which is why the next scale comes into the picture.

Interval Data and Analysis

All the techniques applicable to nominal and ordinal data analysis are applicable to Interval Data as well. Apart from those techniques, there are a few analysis methods such as descriptive statistics, correlation regression analysis which is extensively for analyzing interval data.

Descriptive analysis statistics is the term given to the analysis of numerical data which helps to describe, depict, or summarize data in a meaningful manner and it helps in calculation of mean, median, and mode.

Interval Scale Examples

- There are situations where attitude scales are considered to be interval scales.
- Apart from the temperature scale, time is also a very common example of an interval scale as the values are already established, constant, and measurable.
- Calendar years and time also fall under this category of measurement scales.
- Likert scale, Net Promoter Score, Semantic Differential Scale, Bipolar Matrix Table, etc. are the most-used interval scale examples.

The following questions fall under the Interval Scale category:

- What is your family income?
- What is the temperature in your city?

Ratio Scale: 4th Level of Measurement

Ratio Scale is defined as a variable measurement scale that not only produces the order of variables but also makes the difference between variables known along with information on the value of true zero. It is calculated by assuming that the variables have an option for zero, the difference between the two variables is the same and there is a specific order between the options.

With the option of true zero, varied inferential, and descriptive analysis techniques can be applied to the variables. In addition to the fact that the ratio scale does everything that a nominal, ordinal, and interval scale can do, it can also establish the value of absolute zero. The best examples of ratio scales are weight and height. In market research, a ratio scale is used to calculate market share, annual sales, the price of an upcoming product, the number of consumers, etc.

- Ratio scale provides the most detailed information as researchers and statisticians can calculate the central tendency using statistical techniques such as mean, median, mode, and methods such as geometric mean, the coefficient of variation, or harmonic mean can also be used on this scale.
- Ratio scale accommodates the characteristic of three other variable measurement scales, i.e. labeling the variables, the significance of the order of variables, and a calculable difference between variables (which are usually equidistant).
- Because of the existence of true zero value, the ratio scale doesn't have negative values.
- To decide when to use a ratio scale, the researcher must observe whether the variables have all the characteristics of an interval scale along with the presence of the absolute zero value.
- Mean, mode and median can be calculated using the ratio scale.

Ratio Data and Analysis

At a fundamental level, Ratio scale data is quantitative in nature due to which all quantitative analysis techniques such as SWOT, TURF, Cross-tabulation, Conjoint, etc. can be used to calculate ratio data. While some techniques such as SWOT and TURF will analyze ratio data in such a manner that researchers can create roadmaps of how to improve products or services and Cross-tabulation will be useful in understanding whether new features will be helpful to the target market or not.

Ratio Scale Examples

The following questions fall under the Ratio Scale category:

- What is your daughter's current height?
 - Less than 5 feet.
 - 5 feet 1 inch – 5 feet 5 inches
 - 5 feet 6 inches- 6 feet
 - More than 6 feet
- What is your weight in kilograms?
 - Less than 50 kilograms
 - 51- 70 kilograms
 - 71- 90 kilograms
 - 91-110 kilograms
 - More than 110 kilograms

Attitude measurement scale - Likert Scale

Likert Scale Survey Question

Even those new to survey research have been exposed to the ubiquitous Likert scale. What is a Likert Scale Survey Question?

Likert scale survey question and the strongly agree to strongly disagree framework is a staple in the pantry of questions we regularly use on surveys. In the years I have been involved in constructing surveys and teaching market research I have seen many techniques and question types rise and fall in favor, yet the Likert scale holds its own.

Organizational psychologist Rensis Likert developed the measure in order to assess the level of agreement or disagreement of a symmetric agree-disagree scale. In general, a series of statements each designed to view a construct from a slightly different perspective are leveraged. The power of

this technique is that it works across disciplines – it is just as applicable to a social science construct as it is a marketing one.

For example:

Ted's Pizza has excellent customer service. You get a lot of food for the price at Ted's Pizza. People come to me for information on new products. I enjoy sharing information with others. I consider myself knowledgeable on a variety of issues.

The first two statements measure the customer's perceptions about the business. Qualitative research such as focus groups or in-depth interviews can be useful in helping to generate a list of statements. The last three statements were centered on the individual and might be part of an opinion leader or early adopter scale. Just as easily you can create a scale with items touching upon political or social topics, religion, or other important issues of the day.

To what extent do you agree or disagree with the following statements?							
	SD	2	3	4	5	6	SA
I aspire to a better life for myself.	☐	☐	☐	☐	☐	☐	☐
I do not like going into new situations	☐	☐	☐	☐	☐	☐	☐
My future looks better than my past	☐	☐	☐	☐	☐	☐	☐

Likert scales usually have five, seven, or nine points, with five and seven points used most frequently. For example – Strongly Agree, Agree, No opinion, Disagree, and Strongly Disagree. Adding “Somewhat” to both sides creates the sixth and seventh points. The scales are anchored by strongly agree and strongly disagree. There is some research that indicates having the agree side shown first could inflate the scores. This can be tested by alternating the anchor points within a survey wave and comparing scores in the data analysis stage.

Likert scale is designed to measure attitudes and by nature are multi-item. Basic research tells us that multiple-item measures of a construct are inherently more stable and subject to less random variability than single-item measures. How many items is enough? If you are creating a new scale then you should create as many items as possible and let subsequent analysis narrow the field of contenders. This can be done through brainstorming sessions, focus groups or a review of existing literature.

Semantic Differential Scale

semantic differential scale is a survey or questionnaire rating scale that asks people to rate a product, company, brand, or any 'entity' within the frames of a multi-point rating option. These survey answering options are grammatically on opposite adjectives at each end. For example, love-hate, satisfied-unsatisfied, and likely to return-unlikely to return with intermediate options in between.

Surveys or questionnaires using the semantic differential question is the most reliable way to get information on people's emotional attitude towards a topic of interest.

Charles Egerton Osgood, a famous American psychologist, invented the semantic differential scale so that this 'connotative meaning' of emotional attitude towards entities can be recorded and put to good use.

Osgood conducted this research on an extensive database and found that three scales were commonly useful, irrespective of race or culture or difference in language:

- Estimate: Combination similar to good-bad.
- Authority: Pairs on the lines of powerful-weak.
- Activeness: Combos like active-passive.

Researchers can measure a wide variety of subjects using these combinations, like customers' outlooks about an upcoming product launch or employee satisfaction.

What is the semantic differential scale?

The ease-of-understanding and the popularity it comes with it makes it extremely reliable. The data collection is accurate due to the versatility that these survey questions come with.

Researchers use the semantic differential scale questions to ask respondents to rate products, organization, or services with multi-point questions with polar adjectives at the extremes of this scale like likely/ unlikely, happy/sad, loved the service/ hated the service.

Semantic differential scale examples & question types

1. Slider rating scale: Questions that feature a graphical slider give the respondent a more interactive way to answer the semantic differential scale question.

Which of the following categories best describes your last experience purchasing a product or service on our website?

Very Pleasant Somewhat Pleasant Neither Pleasant nor Unpleasant Somewhat Unpleasant Very Unpleasant

My experience was Somewhat Pleasant



2. Non-slider rating scale: The non-slider question uses typical radio buttons for a more traditional survey look and feel. Respondents are more used to answering.

Given your most recent experience, please rate the following question on a scale of 1 to 7.

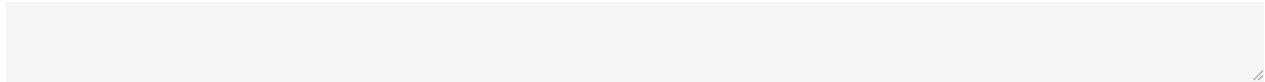
How likely are you to use our service again?

(1) Very Unlikely (2) (3) (4) (5) (6) (7) Very Likely

☐ ☐ ☐ ☐ ☐ ☐ ☐

3. Open-ended questions: These questions give the users ample freedom to express their emotions about your organization, products, or services.

Comments/Suggestions:



4. Ordering: The ordering questions offer the scope to rate the parameters that the respondents feel are best or worst according to their personal experiences.

Based on what you have seen, heard, and experienced, please rank the following brands according to their reliability. Place a "1" next to the brand that is most reliable, a "2" next to the brand that is next most reliable, and so on. Remember, no two cars can have the same ranking.

Ford	-- Select --	▼
BMW	-- Select --	▼
Chrysler	-- Select --	▼
Chevrolet	-- Select --	▼

5. Satisfaction rating: The easiest and eye-catching semantic differential scale questions are the satisfaction rating questions.

How satisfied are you with our services

Very Unsatisfied Unsatisfied Neutral Satisfied Very Satisfied

Thurstone Scale: Definition

Thurstone scale is defined as a unidimensional scale that is used to track respondent's behavior, attitude or feeling towards a subject. This scale consists of statements about a particular issue or topic where each statement has a numerical value that indicates the respondents attitude towards the topic as favorable or unfavorable. Respondents indicate the statements that they agree with, and an average is computed. A mean score of the agreements or disagreements is calculated as the attitude of the respondent towards the topic.

This scale was developed by Robert Thurstone to approximate measurement in equal-appearing interval levels. The Thurstone scale is built upon the fundamentals of the Likert scale, but this method of constructing an attitude scale not only takes value of each item of into consideration while evaluating the final attitude score but also caters to neutral items. The Guttman scale and the Bogardus social distance scale are also variations of unidimensional scales where elements can be ordered in a hierarchical manner.

There are three scales while talking about the Thurstone scale question but the most commonly used method is the equal appearing intervals and hence the scale is also called the equal-appearing interval scale. The two other are slightly more complex to develop but still result in the same agree/disagree quiz question type; they are the **method of successive intervals** and the **method of paired comparisons**.

How to conduct a Thurstone Scale Survey with an Example

An example of a Thurstone scale survey is to understand the attitude of employees in an organization towards diversity hiring in that organization. There are 2 distinct milestones in the Thurstone scale question; to derive the final questions and administer the Thurstone scale question and conduct its analysis.

Derive the Final Question

There are 5 distinctive steps to derive the final question. They are:

Step 1 – Develop statements: Develop a large number of agree/disagree statements on a certain topic. For example, if you wanted to find out people's attitudes towards the policy of diversity hiring in an organization, your statements may include:

- Policy on diversity hiring is wrong.

- Diversity hiring takes away jobs from deserving candidates.
- Diversity hiring brings different viewpoints to the team.
- Diversity hiring brings out the best in the local community.
- Diversity hiring aids in increasing the reputation of a brand

Statement	Median or Mean	IQR
43	1	1.25
21	1	1
16	1.5	1
3	3	2
6	5	2
28	6	1
37	7	3
9	7.5	1.5
18	9	2
26	11	2

Step 2 – Rank each statement with a panel of judges: The next step would be to have a panel of judges rate each item on a scale of 1 to 11 where 1 is the least favorable attitude towards the common vector – diversity hiring and 11 is an extremely favorable attitude. It is important to note that the judges are required to rate each option and not agree or disagree with them.

Step 3 – Calculate median and/or mean and Interquartile range (IQR): The data collected from all judges is to then be analyzed to draw up a table with the mean or median values in the ascending order. Using median or mean is a personal choice and the options

throw up accurate results in the use of either. If you have 50 statements, you need to have 50 means/medians and 50 IQR's.

Step 4 – Sorting the table: The data has to be sorted on the basis of smallest to largest median/mean and the IQR's for each median/mean have to be in a descending order. This can be represented as:

Step 5 – Selecting final variables or options: Select options on the basis of the above table. For example, you could select one item from each mean/median value. You want the statements with the most agreement between judges. For each median value, this is the item with the lowest interquartile range. This is a “Rule of Thumb”: you don't have to choose this item. If you decide it's poorly worded or ambiguous, choose the item above it (with the next lowest IQR).

Administering and Analysis of the Thurstone Scale Question

Once the final questions have been decided upon, they are shared with the respondents to select from agreement or disagreement. The ratings are shown in parenthesis but the rating is not shared with the actual respondents.

- **Administering and analysis by median or mean:** The question and the subsequent options can be administered to the respondents by using either the mean or the median score in the below format. The weights of the statements are summed and divided by the number of statements checked. If a respondent were to agree with statements 2, 5, 7 and 10; the attitude score is $10.5 + 2.5 + 4.5 + 6.0 = 23.5/4 = 5.8$. Dividing the number of statements puts this score at a little above the midway point of the 1-11 scale. This score indicates that the attitude is slightly favorable to diversity hiring in the organization.

Please check all those statements with which you agree.

	Agree	Disagree
1. Policy on diversity hiring is wrong (3.0)	☐	☐
2. Diversity hiring takes away jobs from deserving candidates. (10.5)	☐	☐
3. Diversity hiring brings different viewpoints to the team. (8.5)	☐	☐
4. Diversity hiring brings out the best in the local community. (1.0)	☐	☐
5. Diversity hiring aids in a increasing the reputation of a brand. (2.5)	☐	☐
6. Diversity hiring is detrimental to collaboration (5.5)	☐	☐
7. Such people are generally from the lower rungs of the society. (4.5)	☐	☐
8. Acquired diversity is more important than inherent diversity. (10.5)	☐	☐
9. Diversity hiring increases productivity in teams. (2.0)	☐	☐
10. Diversity hiring leads to interpersonal issues. (6.0)	☐	☐
11. Diversity hiring diminishes discriminatory behavior. (9.5)	☐	☐

- **Administering and analysis by simple count or percentage:** In the same example above, if the question is administered without the mean or median score, the calculation can be depicted by a simple count of agreement on the 1-11 scale or a percentage. If the respondent were to agree with statements 1, 4, 5, 6, 8, 9 and 11; the agreement count is 7 out of 11 which puts the percentage at 63.63% which means that the attitude is favorable towards diversity hiring.

Characteristics of a Thurstone Scale Question

Some distinct characteristics of a Thurstone scale question are:

- **They are two stepped:** A Thurstone scale question is never administered at the first iteration stage without the rating by judges. This is an important characteristic of this question as the options that a respondent sees are weighted and there is consensus of them being in the survey.

- **Mean or median is always calculated:** Since each option is weighted, the mean or median is calculated for each option. This also forms the basis of the selection method for use in the final survey.
- **Only agree or disagree options:** The respondent makes a selection only on the basis of the agreement or disagreement with the statement.

Use of a Thurstone Scale Survey

The Thurstone scale survey is used to measure the respondents attitudes on a particular subject. The scale can be applied to a wide range of market research surveys, in market research including:

- **Surveys that measure opinions:** The Thurstone scale question produces quantifiable data about the measures of strength of the respondents opinions.
- **Those that gauge attitudes or feelings:** This scale is used effectively in customer satisfaction to predict future purchasing trends and in employee engagement to calculate turnover.

Multi-Dimensional Scaling: Non comparative scaling techniques

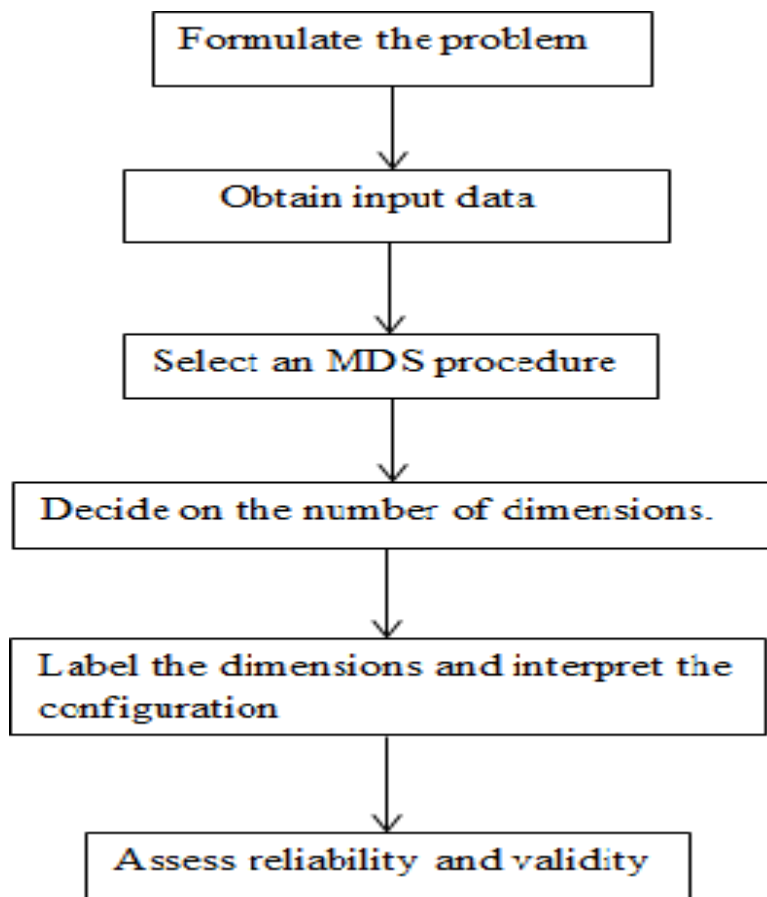
Multidimensional Scaling (MDS) is a class of procedures for representing perception and preferences of respondents spatially by means of a visual display. Perceived or psychological relationships among points in a multidimensional space. These geometric representations are often called spatial maps. The axes of the spatial map are assumed to denote the psychological bases or underlying dimensions respondents use to form perceptions and preferences for stimuli. MDS has been used in marketing to identify:

1. The number and nature of dimensions consumers use to perceive different brands in the marketplace
2. The positioning of current brands on these dimensions
3. The positioning of consumers ideal brand on these dimensions

Information by MDS has been used for a variety of marketing applications, including:

- i. Image measurement. Compare the customers' and noncustomers' perceptions of the firm with the firm's perceptions of itself and thus identify perceptual gaps.
- ii. Market segmentation. Position brands and consumers in the same space and thus identify groups of consumers with relatively homogeneous perceptions.
- iii. New product development. To look for gaps in the spatial map, which indicate potential opportunities for positioning new products. Also, to evaluate new product concepts and existing brands on attest basis to determine how consumers perceive the new concepts. The proportion of preferences for each new product is one indicator of its success.

- iv. Assessing advertising effectiveness. Spatial maps can be used to determine whether advertising has been successful in achieving the desired brand positioning.
- v. Pricing analysis. Spatial maps developed with and without pricing information can be compared to determine the impact of pricing.
- vi. Channel decisions. Judgments on compatibility of brands with different retail outlets could lead to spatial maps useful for making channel decisions.
- vii. Attitude scale construction. MDS techniques can be used to develop the appropriate dimensionality and configuration of the attitude space.



Formulate the problem

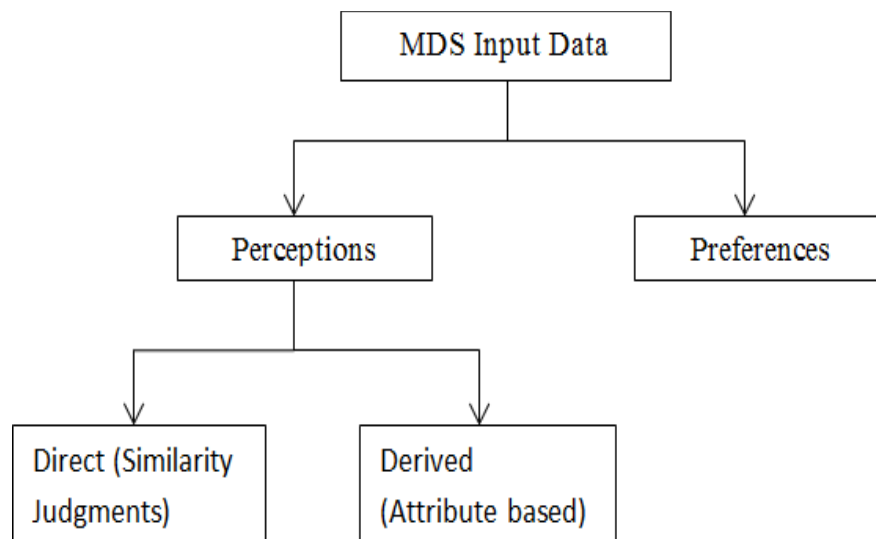
Formulating the problem requires that the researcher specify the purpose for which the MDS results would be used and select the brand or other stimuli to be included in analysis. The number of brand or stimuli selected and the specific brands included determine the nature of the resulting dimensions and configurations. At a minimum, eight brands or stimuli should be included so as to

obtain a well-defined spatial map. Including more than 25 brands is likely to be cumbersome and may result in respondent fatigue.

The decision regarding which specific brands or stimuli to include should be made carefully. Suppose a researcher is interested in obtaining consumer perceptions of automobiles. If luxury automobiles are not included in the stimuli set, this dimension may not emerge in the results. The choice of number and specific brand and stimuli to be included should be based on the statement of marketing research problem, theory and the judgments of researcher.

Obtain input data

As shown Figure 2, input data obtained from the respondents may be related to perception or preferences. Perception data, which may be direct or derived, is discussed first.



Input data for Multi-dimensional Scaling

Perception data: Direct Approaches.

In direct approaches to gathering perception data, the respondents are asked to judge how similar or dissimilar the various brands or stimuli are using their own criteria. Respondents are often required to rate all possible pair of brands or stimuli in terms of similarity on a Likert scale. These data are referred to as similarity judgments. For example, similarity judgment on all possible pairs of toothpaste brands may be obtained in the following manner.

Perception Data : Derived Approaches, Derived approaches to collect perception data are attribute-based approaches requiring the respondent to rate the brand or stimuli on the identified attributes

using semantic differential or Likert scale example, the different brands of toothpastes may be rated on attributes such as these

Sometimes an ideal brand is also included in the stimulus set. The respondent asked to evaluate their hypothetical ideal brand on the same set of attribute. If attribute rating are obtained, a similarity measure (such as Euclidean distance) is derived for pair of brand.

Direct vs. Derived Approaches:

Direct approaches have advantages thus researcher does not have to identify a set of salient attributes. Respondent makes stimuli judgment using their own criteria, as they would under normal circumstances. Disadvantages is that the criteria are influenced by the brands and stimuli being evaluated. If the various brands of automobiles being evaluated are in the same price range, then price will not emerge not as an important factor. It may be difficult to determine before analyze and how the individual respondent judgment should be combined. Furthermore, it is difficult to label the dimension of the spatial map.

The advantage of the attribute-based approach is that it is easy to identify respondent with homogeneous perceptions. The respondent can be clustered the advantage of the dimension based on rating. It is also easier to label the dimension. A disadvantage is that

the researcher identifies all salient attributes, a different task. The spatial map obtained depends on the attributes identified.

Preference data: Preference data order the brand or stimuli in terms of respondent preference for some property. A common way in which such data are obtained is preference rating. Respondents are required to rank the brand from the most preferred to the least preferred. Alternatively, respondent may be required to make paired comparison and indicate which brand in a pair they prefer. Another method is to obtain preference rating for the various brands. When spatial maps are based on preference data, distance implies difference in preference. The configuration derived from preference data may differ greatly from that obtained from similarity data. Two brands may be perceived as different in similarity map yet similar in a preference map and vice versa. For example, Crest and Pepsodent may be perceived by a group of respondents as very different brands and thus appear far apart on a perception map. However these two brands may be about equally preferred and appear close together on a preference map.

Select an MDS Procedure

Selection of a specific MDS procedure depends upon whether perception or preference data are being scaled, or whether the analysis requires both kinds of data. The nature of the input data is also a determining factor. Nonmetric MDS procedures assume that the input data are ordinal, but they result in the metric output. The distances in the resulting spatial map may be assumed to be interval scaled. These procedures find, in a given dimensionality, a spatial map whose rank orders of

estimated distances brands and stimuli best preserve or reproduce the input rank orders. In contrast, Metric MDS methods assume that input data are metric. Because the output is also metric, a stronger relationship between the output and input data is maintained, and the metric (interval or ratio) qualities of the input data are preserved. The metric and non-metric methods produce similar results.

Another factor influencing the selection of a procedure is whether the MDS analysis will be conducted at the individual respondent level or at an aggregate level. In individual level analysis, the data is analyzed separately for each respondent, resulting in a spatial map for each respondent. Although individual level analysis is useful from a research perspective, it is not appealing from managerial standpoint. Marketing strategies are typically formulated at the segment or aggregate level, rather than at the individual level. If the aggregate-level analysis is conducted, some assumption must be made in aggregating individual data. Typically it is assumed that all respondents use the same dimensions to evaluate the brand or stimuli, but that different respondents weight these common dimensions differentially.

Label the Dimensions and Interpret the Configuration

Once a spatial map is developed, the dimensions must be labeled and the configuration interpreted. Labeling the dimensions requires subjective judgment on the part of the researcher. The following guidelines can assist in this task:

1. Even if direct similarity judgments are obtained, ratings of the brands or researcher supplied attributes may still be collected. The axes may then be labeled for the attributes with which they are most closely aligned.
2. After providing direct similarities or preference data, the respondents may be asked to indicate the criteria used in making their evaluations. These criteria may then be subjectively related to the spatial map to label the dimensions.
3. If possible, the respondents can be shown their spatial maps and asked to label the dimensions by inspecting the configurations.
4. If objective characteristics of the brands are available (e.g., horsepower or miles per gallon for automobiles), these could be used as an aid in interpreting the subjective dimensions of spatial maps.

Assess Reliabilities and Validity

The input data, and consequently the MDS solutions, are invariably subject to sub random variability. Hence, it is necessary that some assessment be made of the relation and validity of MDS solutions. The following guidelines are suggested.

1. The index of fit, or R -square, should be examined. This is squared correlation index that indicates the proportional of variance of the optimally scaled data that can be accounted for the MDS procedure. Thus, it indicates how well the MDS model fits the input data. Although higher values of R -square are desirable, values of 0.60 or better are considered acceptable.

2. Stress values are also indicative of the quality of MDS solutions. Whereas *R*-square is a measure of goodness of fit, stress measures badness of fit, or the proportion of variance of the optimally scaled data that is not accounted for the MDS model. Stress values vary with the type of MDS procedure and the data being analyzed. For Kruskal's stress formula 1, the recommendations for evaluating stress values are as follows.

Stress (%)	Goodness of fit
20	poor
10	fair
5	good
2.5	excellent
0	perfect

3. If an aggregate-level analysis has been done, the original data should be split into two or more parts. MDS analysis should be conducted separately on each part and the results compared.
4. Stimuli can be selectively from the input data and the solutions determined for the remaining stimuli.
5. A random error term could be added to the input data. The resulting data are subjected to MDS analysis and the solutions compared.
6. The input data could be collected at two different points in time and the test-retest reliability different.

Assumption and limitations of MDS

It is worthwhile to point out some assumptions and limitations of MDS. It is assumed that the similarity of stimulus A to B is the same as the similarity of stimulus B to A there are some instances where the assumption may be violated. For example, Mexico is perceived as more similar to United States than the United States is to Mexico. MDS assumes that the distance (similarity) between two stimuli is some function of their partial similarities on each of several perceptual dimensions. Not much research has been done to test this assumption. When a spatial map is obtained, it is assured that inter point distances are ratio scaled and that the axes of the map are multi dimensionally interval scaled. A limitation of MDS is that dimension interpretation relating physical changes in brands or stimuli to changes in the perceptual map is difficult at best.
